

Analyzing Hallucination Patterns in RAG-based Financial Question Answering

A controlled study on FinanceBench
(150 QA · 84 SEC filings)

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In finance, there are no minor hallucinations

Open-domain

- Plausible but slightly off
- Low stakes
- Easy to correct

Financial (zero-tolerance)

- **One wrong figure**
- **Confidently wrong**
- **High stakes**

Model output: ~~\$25.8 billion~~
**SEC filing dictates:
\$3.2 billion**

Why FinanceBench is hard

Long, tabular context

128 pages

(median per filing)

69% of evidence is table-heavy.

Short, numeric answers

9 words

(median gold answer)

84% numeric.

Abstractive

2% verbatim

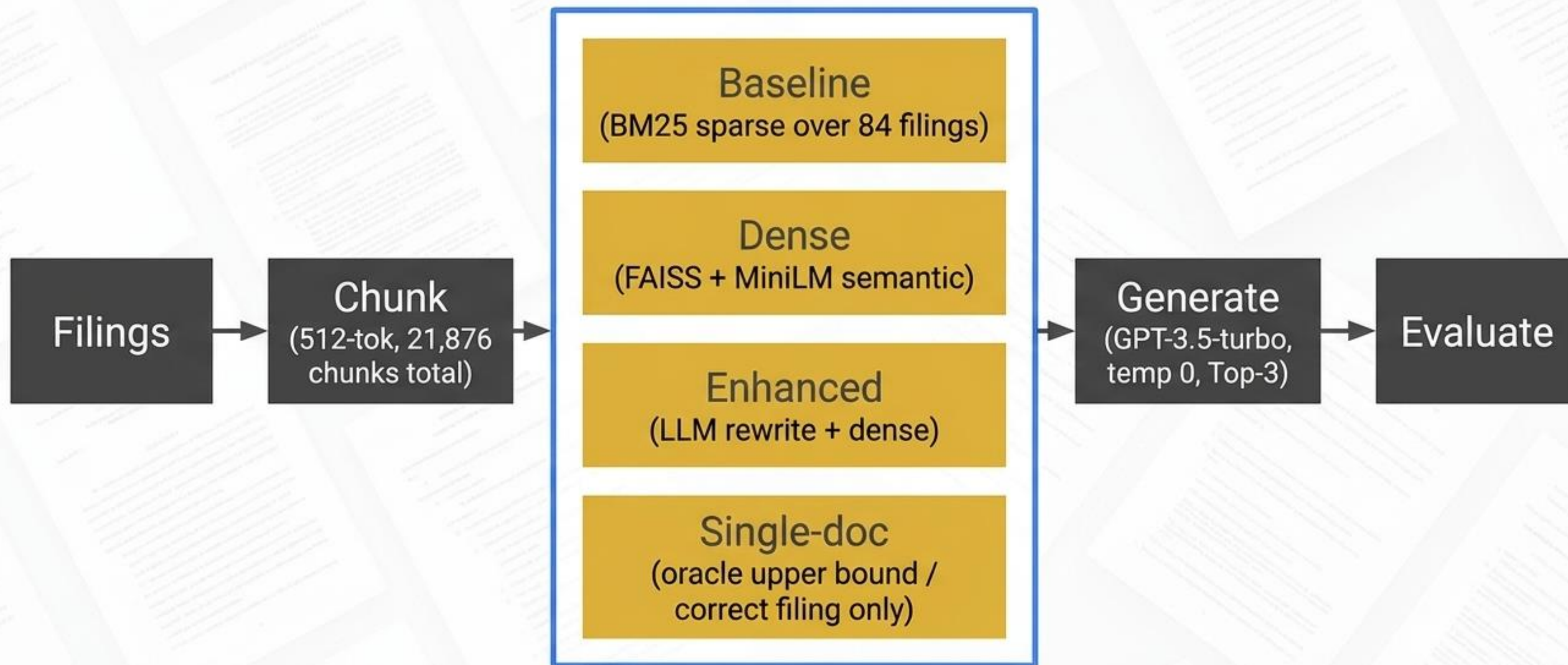
(only 2% of gold answers appear word-for-word in the evidence).

Modest lexical overlap

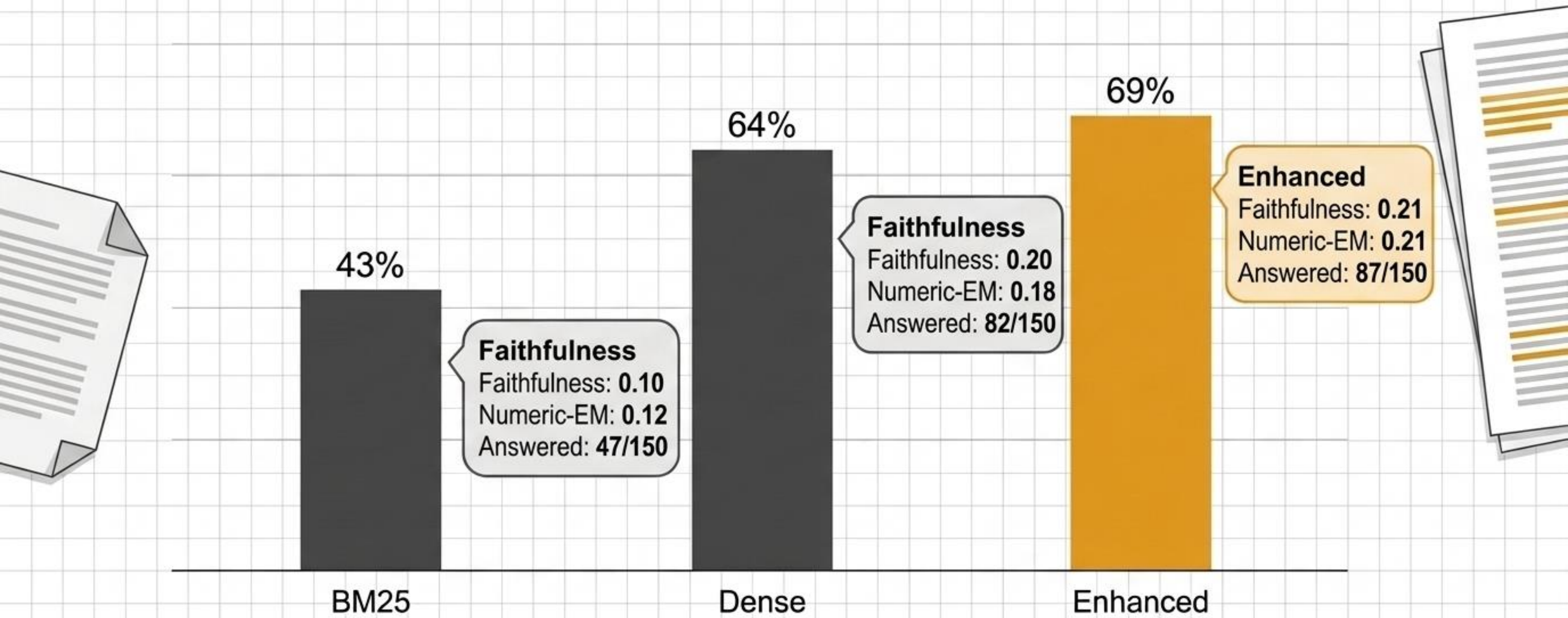
0.47 cosine

(median question–evidence overlap)
46% require computation.

Method: the pipeline showdown

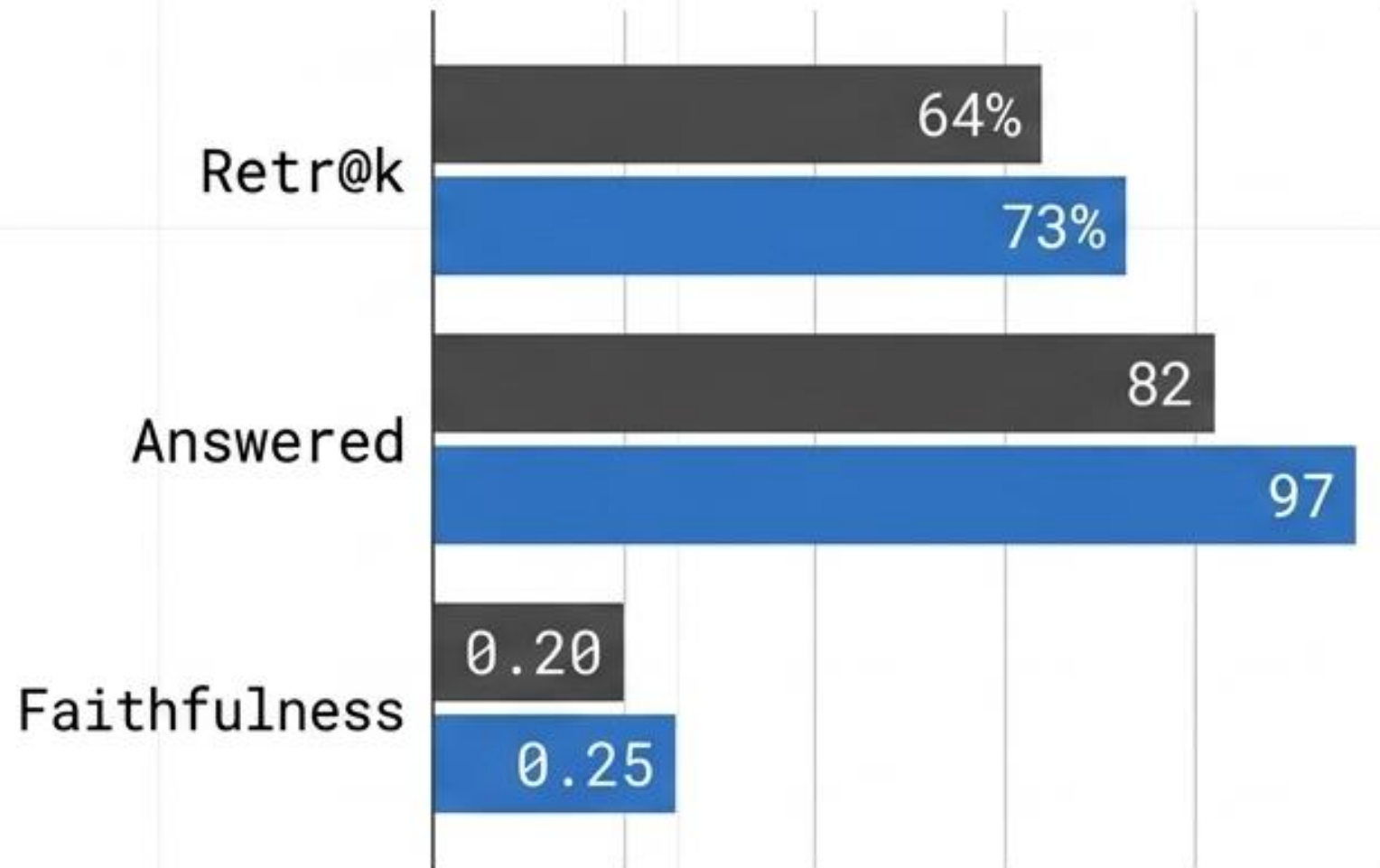


RQ1: retrieval is the dominant lever

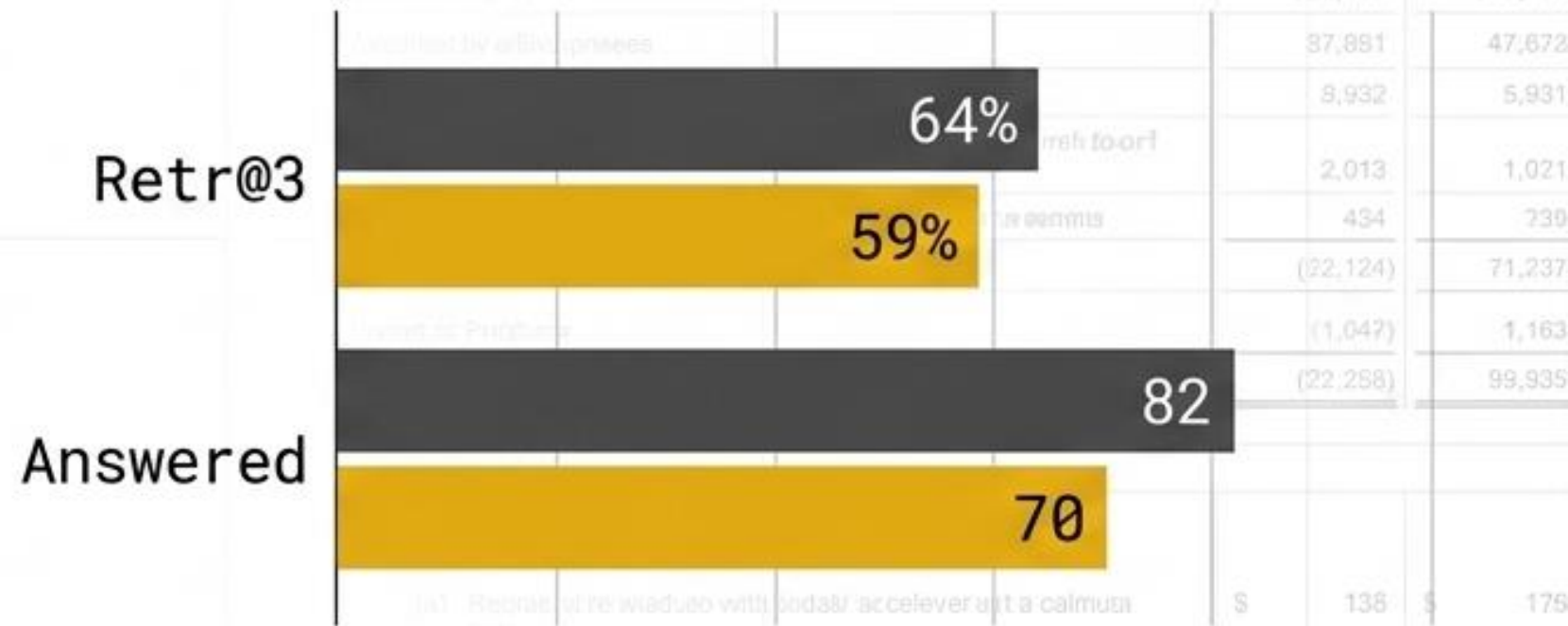


Ablations: depth and chunk size

A Top-3 vs Top-5

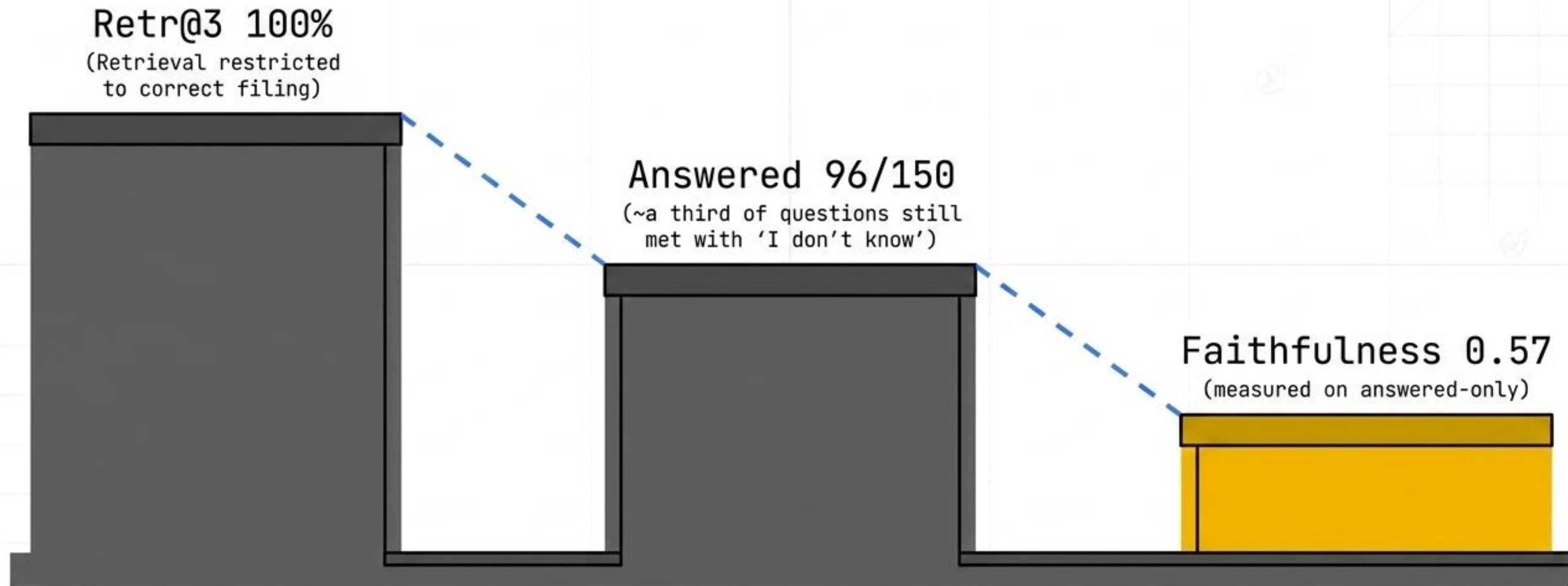


B 512-token vs 256-token



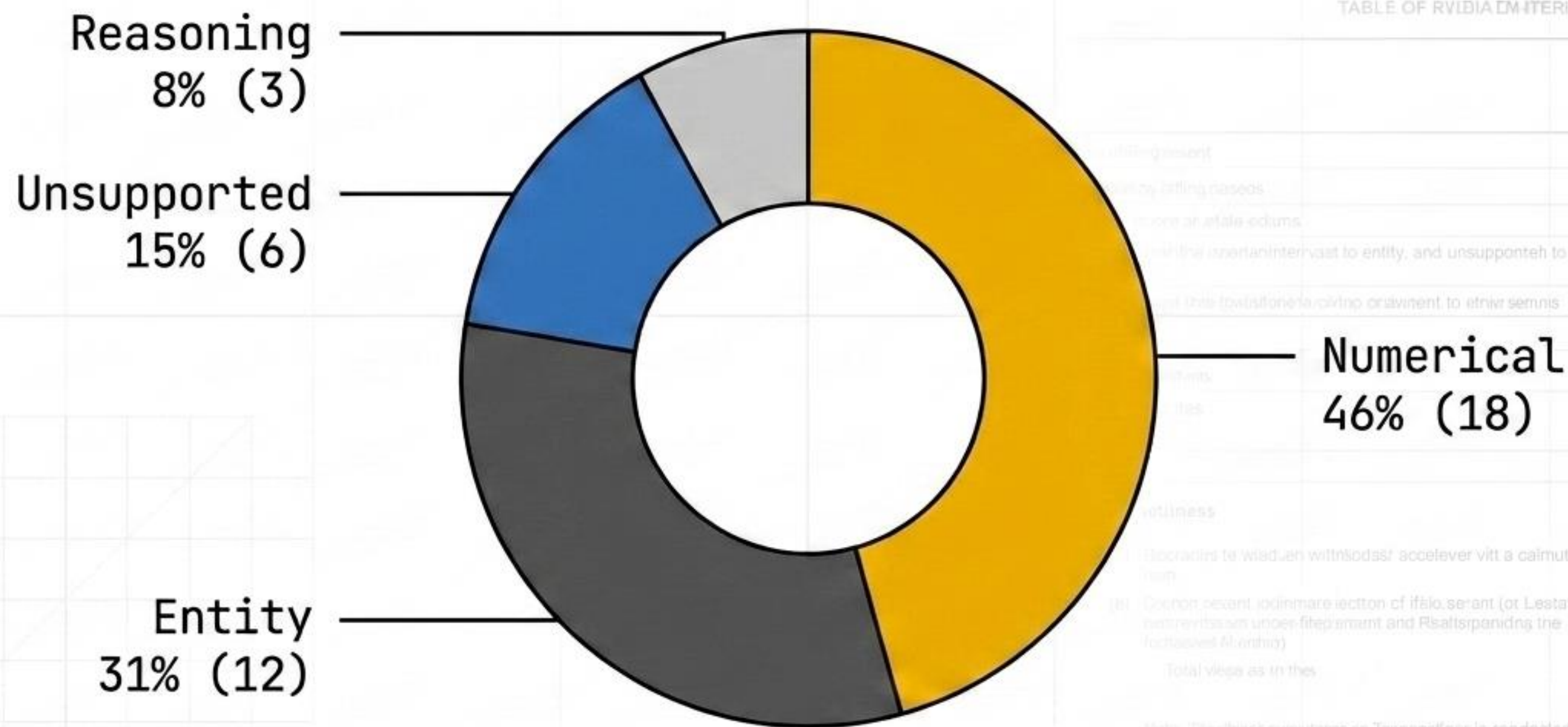
WARNING: 256-token chunks fail by splitting numeric figures from their labels in SEC tables.

The single-document ceiling



Takeaway: Finding the right filing is necessary but not sufficient.

RQ2: an autopsy of the hallucinations



Note: Hand-labeled the 39 genuine hallucinations out of the answered-but-incorrect cases.

Case study: the entity error

Scenario: What drove AMD's FY22 revenue change?



Takeaway: The model faithfully used the wrong document — entity errors are retrieval collisions in disguise.

The evaluation trap: when the metric lies

SEC DOCUMENT #: [REDACTED]
REDACTED BY IND.SG26R1J17LY OF SEC
DOCUMENTS WATERISTED SK1 NRC RELOWZED
DOCUMENTS: [REDACTED]

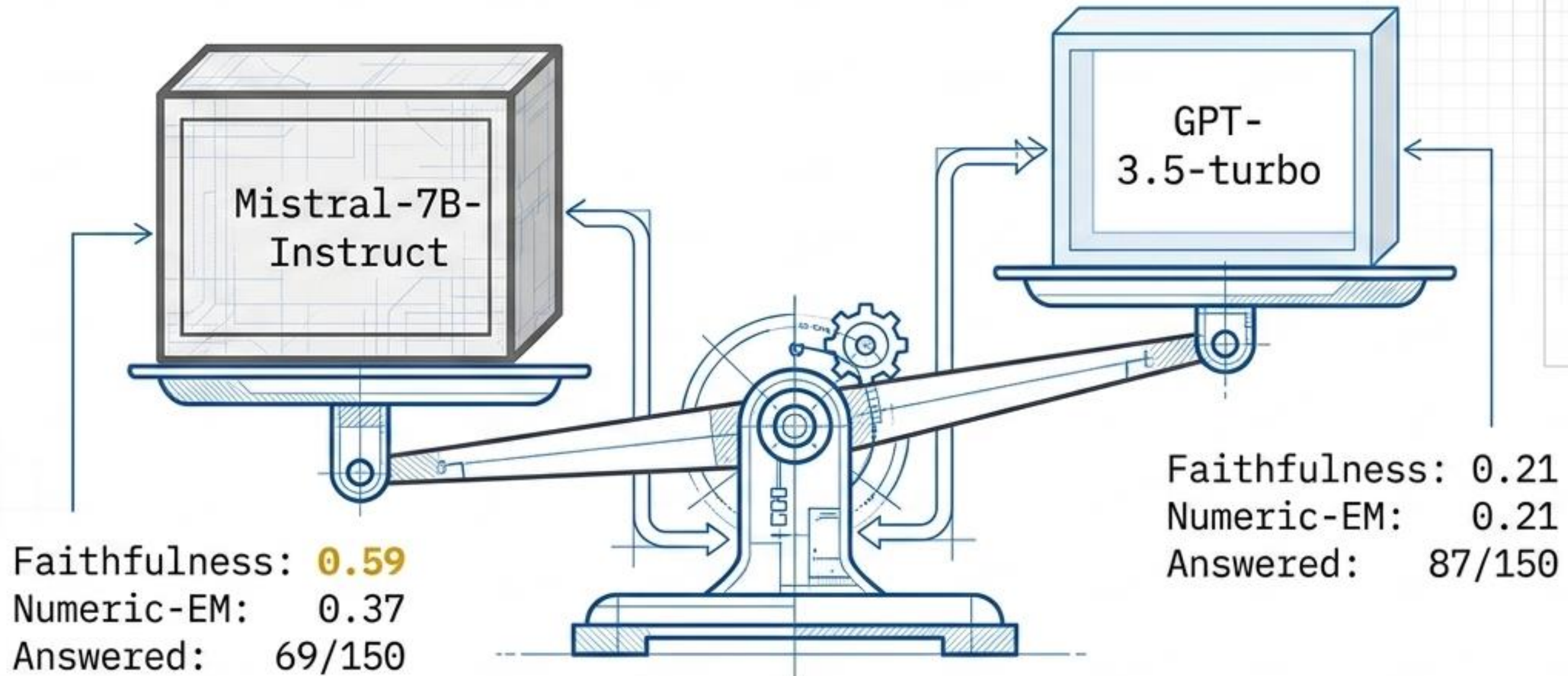
Model (\$1,577 million) ~~=~~ Gold (\$1.577 billion)

Exact Match 0

- ~22% (11 of 50) of automatically flagged hallucinations were actually correct.
- Standard exact-match metrics carry a massive false-positive rate on financial formatting. Numeric-tolerant EM is mandatory.

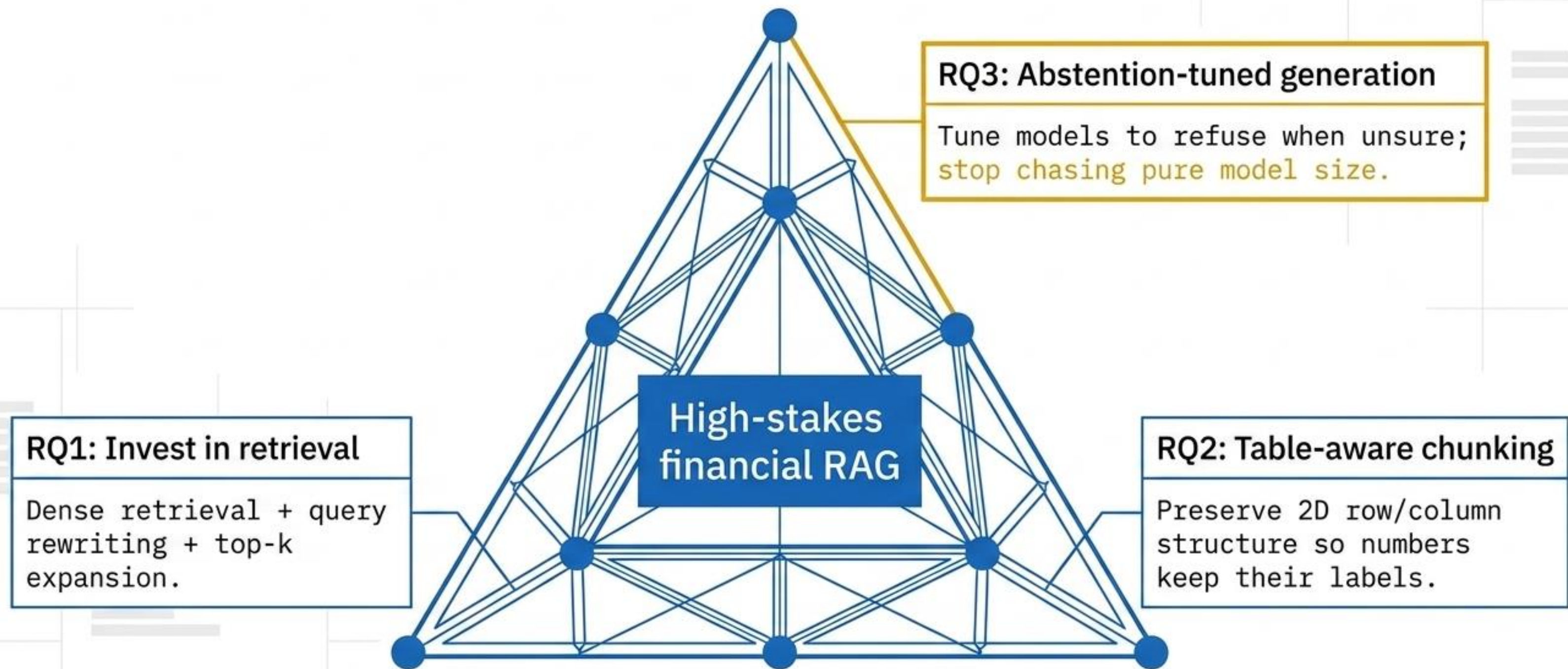
REDACTED WITH: [REDACTED]
SEC DOCUMENTS: [REDACTED]

RQ3: the abstention paradox



Takeaway: The smaller model is more faithful because it refuses when unsure. Abstention behavior, not model scale, drives grounding.

The RAG architect's blueprint



Conclusion

Financial hallucinations are primarily retrieval collisions and table-parsing failures, not free invention.

Automatic exact-match metrics are fundamentally unreliable for numeric grounding.

The most robust pipelines combine strong retrieval with abstention-tuned generation.

Limitations: Single annotator; 150-question subset; PDF extraction flattens tables.

Bonus Insight: A RoBERTa detector trained on RAGTruth transfers to this domain at an F1 of 0.81, but over-flags.

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DATASET: FINANCEBENCH

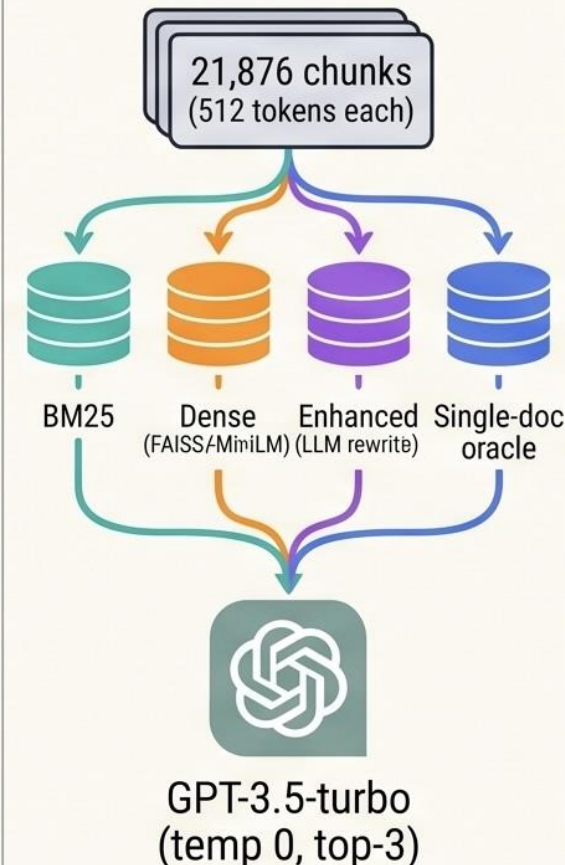


150 questions, 84 SEC filings, 32 companies



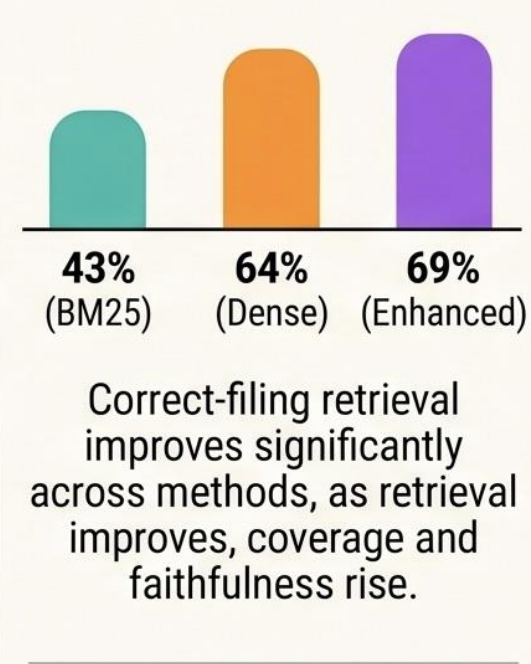
The dataset is highly challenging: **84%** of answers are numeric, and **98%** must be synthesized (only 2% are verbatim).

METHOD: THE PIPELINE



RAGAS judge = GPT-4o-mini

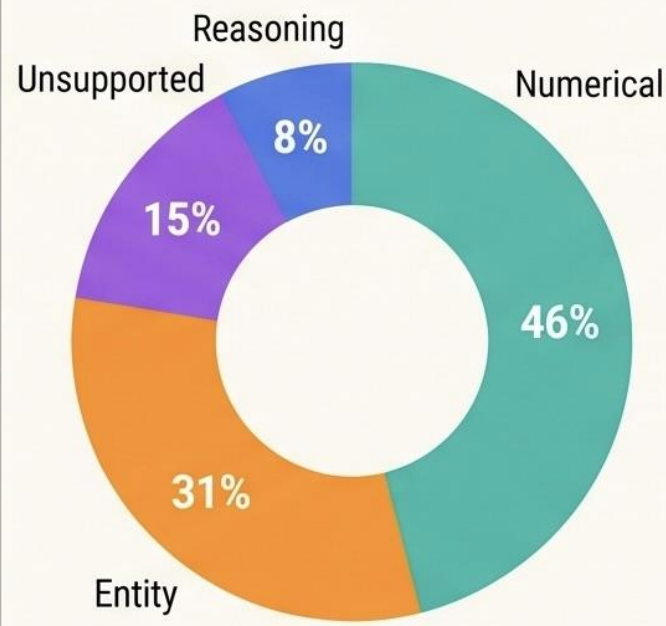
RQ1: RETRIEVAL IS THE DOMINANT LEVER



Top-k Performance: Top-3 vs Top-5
Top-3 has **64%** recall and 0.20 faithfulness;
Top-5 has **75%** recall and 0.25 faithfulness.

Celling: 100% retrieval but only 96/150 answered
Even with perfect retrieval, the generator struggles to answer nearly one-third of the questions.

RQ2: TAXONOMY OF 39 GENUINE HALLUCINATIONS



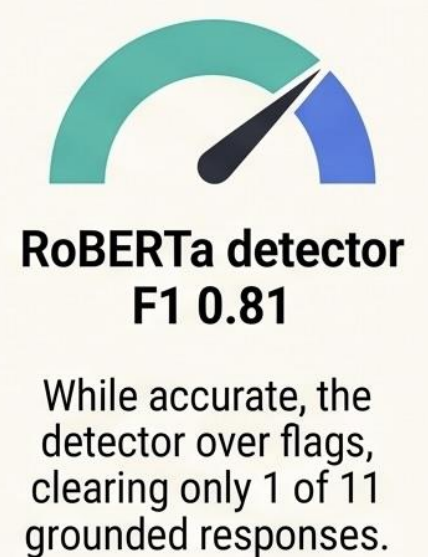
~22% of flagged failures were actually correct
These “metric false positives” occur when the model provides the correct value in a different format (e.g., \$1,577 million vs \$1.577 billion).

RQ3: GENERATOR COMPARISON & THE ABSTENTION PARADOX



Abstention drives grounding
Mistral is more faithful because it abstains more, it refuses to answer when unsure, whereas GPT-3.5 attempts to answer and hallucinates.

DETECTION: PERFORMANCE & ERROR ANALYSIS



	Predicted Grounded	Predicted Hallucinated
Actual Grounded	1	10
Actual Hallucinated	6	33

